

Exercise 1: Setting Up Junit

```
src > main > java > com > example > Calculator.java > {} com.example
1  package com.example;
2
3  public class Calculator {
4      public int add(int a, int b) {
5          return a + b;
6      }
7
8      public int subtract(int a, int b) {
9          return a - b;
10     }
11
12     public int multiply(int a, int b) {
13         return a * b;
14     }
15
16     public double divide(int a, int b) {
17         if (b == 0) {
18             throw new IllegalArgumentException(s: "Cannot divide by zero");
19         }
20         return (double) a / b;
21     }
22 }
23
```

```
src > test > java > com > example > J CalculatorTest.java > {} com.example
```

```
1 package com.example;
2
3 import org.junit.Test;
4 import static org.junit.Assert.assertEquals;
5
6 public class CalculatorTest {
7
8     private final Calculator calculator = new Calculator();
9
10    @Test
11    public void testAdd() {
12        assertEquals(5, calculator.add(a: 2, b: 3));
13        assertEquals(-1, calculator.add(-2, b: 1));
14        assertEquals(0, calculator.add(a: 0, b: 0));
15    }
16
17    @Test
18    public void testSubtract() {
19        assertEquals(1, calculator.subtract(a: 3, b: 2));
20        assertEquals(-3, calculator.subtract(-2, b: 1));
21        assertEquals(0, calculator.subtract(a: 5, b: 5));
22    }
23
24    @Test
25    public void testMultiply() {
26        assertEquals(6, calculator.multiply(a: 2, b: 3));
27        assertEquals(-2, calculator.multiply(-2, b: 1));
28        assertEquals(0, calculator.multiply(a: 0, b: 5));
29    }
30
31    @Test
32    public void testDivide() {
33        assertEquals(2.0, calculator.divide(a: 6, b: 3), 0.0001);
34        assertEquals(-2.0, calculator.divide(-6, b: 3), 0.0001);
35    }
36
37    @Test(expected = IllegalArgumentException.class)
38    public void testDivideByZero() {
39        calculator.divide(a: 5, b: 0);
40    }
41 }
42
```

```
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.CalculatorTest
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.181 s -- in com.example.CalculatorTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 5.991 s
[INFO] Finished at: 2026-07-03T12:30:46+05:30
[INFO] -----
```

Exercise 3: Assertions in Junit

s D:\LOCAL\FSE WEEKWISE\junit-setup\src\test\java\com\example\AssertionTest.java

```
1  import static org.junit.Assert.*;
2  import org.junit.Test;
3
4  public class AssertionsTest {
5
6      @Test
7      public void testAssertions() {
8
9          // Assert equals
10         assertEquals(5, 2 + 3);
11
12         // Assert true
13         assertTrue(5 > 3);
14
15         // Assert false
16         assertFalse(5 < 3);
17
18         // Assert null
19         assertNull(null);
20
21         // Assert not null
22         assertNotNull(new Object());
23     }
24 }
```

```
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.CalculatorTest
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.181 s -- in com.example.CalculatorTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 5.991 s
[INFO] Finished at: 2026-07-03T12:30:46+05:30
[INFO] -----
```

Exercise 1: Mocking and Stubbing

s D:\LOCAL\FSE WEEKWISE\junit-setup\src\test\java\com\example\AssertionTest.java

```
1  import static org.junit.Assert.*;
2  import org.junit.Test;
3
4  public class AssertionsTest {
5
6      @Test
7      public void testAssertions() {
8
9          // Assert equals
10         assertEquals(5, 2 + 3);
11
12         // Assert true
13         assertTrue(5 > 3);
14
15         // Assert false
16         assertFalse(5 < 3);
17
18         // Assert null
19         assertNull(null);
20
21         // Assert not null
22         assertNotNull(new Object());
23     }
24 }
```

src > main > java > com > example > J ExternalApi.java > ExternalApi

```
1  package com.example;
2
3  public interface ExternalApi {
4
5      String getData();
6
7  }
```

src > main > java > com > example > J MyServices.java > MyService

```
1 package com.example;
2
3 public class MyService {
4
5     private ExternalApi api;
6
7     public MyService(ExternalApi api) {
8         this.api = api;
9     }
10
11     public String fetchData() {
12         return api.getData();
13     }
14 }
```

src > test > java > com > example > J MyServicesTest.java > MyServiceTest

```
1 package com.example;
2
3 import static org.junit.jupiter.api.Assertions.assertEquals;
4 import static org.mockito.Mockito.*;
5
6 import org.junit.jupiter.api.Test;
7
8 public class MyServiceTest {
9
10     @Test
11     public void testExternalApi() {
12
13         // Create mock object
14         ExternalApi mockApi = mock(ExternalApi.class);
15
16         // Stub method
17         when(mockApi.getData()).thenReturn("Mock Data");
18
19         // Create service
20         MyService service = new MyService(mockApi);
21
22         // Call service
23         String result = service.fetchData();
24
25         // Verify result
26         assertEquals("Mock Data", result);
27     }
28 }
```

Exercise 2: Verifying Interactions

s D:\LOCAL\FSE WEEKWISE\junit-setup\src\test\java\com\example\AssertionTest.java

```
1  import static org.junit.Assert.*;
2  import org.junit.Test;
3
4  public class AssertionsTest {
5
6      @Test
7      public void testAssertions() {
8
9          // Assert equals
10         assertEquals(5, 2 + 3);
11
12         // Assert true
13         assertTrue(5 > 3);
14
15         // Assert false
16         assertFalse(5 < 3);
17
18         // Assert null
19         assertNull(null);
20
21         // Assert not null
22         assertNotNull(new Object());
23     }
24 }
```

src > main > java > com > example > J ExternalApi.java > ExternalApi

```
1  package com.example;
2
3  public interface ExternalApi {
4
5      String getData();
6
7  }
```

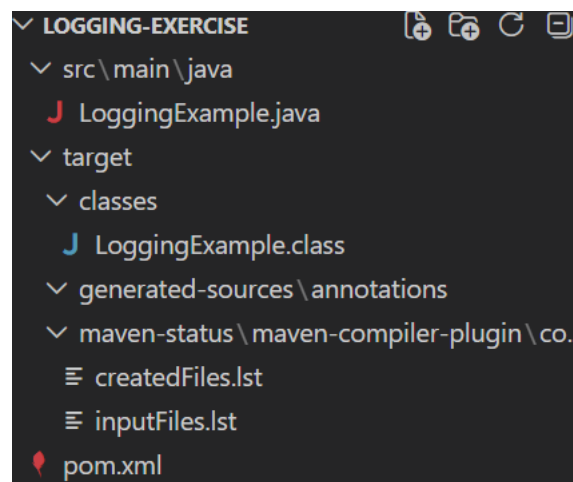
src > main > java > com > example > J MyServices.java > MyService

```
1 package com.example;
2
3 public class MyService {
4
5     private ExternalApi api;
6
7     public MyService(ExternalApi api) {
8         this.api = api;
9     }
10
11     public String fetchData() {
12         return api.getData();
13     }
14 }
```

src > test > java > com > example > J MyServicesTest.java > MyServiceTest

```
1 package com.example;
2
3 import static org.junit.jupiter.api.Assertions.assertEquals;
4 import static org.mockito.Mockito.*;
5
6 import org.junit.jupiter.api.Test;
7
8 public class MyServiceTest {
9
10     @Test
11     public void testExternalApi() {
12
13         // Create mock object
14         ExternalApi mockApi = mock(ExternalApi.class);
15
16         // Stub method
17         when(mockApi.getData()).thenReturn("Mock Data");
18
19         // Create service
20         MyService service = new MyService(mockApi);
21
22         // Call service
23         String result = service.fetchData();
24
25         // Verify result
26         assertEquals("Mock Data", result);
27     }
28 }
```

Exercise 1: Logging Error Messages and Warning Levels



```
src > main > java > J LoggingExample.java > ...
1  import org.slf4j.Logger;
2  import org.slf4j.LoggerFactory;
3
4  public class LoggingExample {
5      private static final Logger logger = LoggerFactory.getLogger(LoggingExample.class);
6
7      public static void main(String[] args) {
8          logger.error("This is an error message");
9          logger.warn("This is a warning message");
10     }
11 }
12
```



```
pom.xml > {} Grammars > http://maven.apache.org/xsd/maven-4.0.0.xsd > Cache > file:///C:/Users/prane/.lemminx/cache/http://maven.apache.org/xsd/maven-4.0.0.xsd
1 <project xmlns="http://maven.apache.org/POM/4.0.0"
2   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
4   <modelVersion>4.0.0</modelVersion>
5
6   <groupId>com.example</groupId>
7   <artifactId>logging-exercise</artifactId>
8   <version>1.0-SNAPSHOT</version>
9
10  <properties>
11    <maven.compiler.source>11</maven.compiler.source>
12    <maven.compiler.target>11</maven.compiler.target>
13    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
14  </properties>
15
16  <dependencies>
17    <dependency>
18      <groupId>org.slf4j</groupId>
19      <artifactId>slf4j-api</artifactId>
20      <version>1.7.30</version>
21    </dependency>
22    <dependency>
23      <groupId>ch.qos.logback</groupId>
24      <artifactId>logback-classic</artifactId>
25      <version>1.2.3</version>
26    </dependency>
27  </dependencies>
28
29  <build>
30    <plugins>
31      <plugin>
32        <groupId>org.codehaus.mojo</groupId>
33        <artifactId>exec-maven-plugin</artifactId>
34        <version>3.0.0</version>
35        <configuration>
36          <mainClass>LoggingExample</mainClass>
37        </configuration>
38      </plugin>
39    </plugins>
40  </build>
41 </project>
42
```